📍📍 Assignment: Subsetting and Modifying Data📍📍

`READ THE BIG MART SALES DATA`

In [1]:
```
import pandas as pd
```

In [2]:
```
data = pd.read_csv('dataset/big_mart_sales.csv')
```

In [3]:
```
data.head()
```

Out[3]:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Loca
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	OUT049	1999	Medium	
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	OUT018	2009	Medium	
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	OUT049	1999	Medium	
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	OUT010	1998	NaN	
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	OUT013	1987	High	

`Select the rows where Item_MRP is less than 100`

In [4]:
```
# WRITE YOUR CODE HERE
data[data.Item_MRP < 100]
```

Out[4]:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_l
1	DRC01	5.920	Regular	0.019278	Soft Drinks	48.2692	OUT018	2009	Medium	
4	NCD19	8.930	Low Fat	0.000000	Household	53.8614	OUT013	1987	High	
5	FDP36	10.395	Regular	0.000000	Baking Goods	51.4008	OUT018	2009	Medium	
6	FDO10	13.650	Regular	0.012741	Snack Foods	57.6588	OUT013	1987	High	
8	FDH17	16.200	Regular	0.016687	Frozen Foods	96.9726	OUT045	2002	NaN	
...	...	...	...	...	...	...	...	...	...	
8513	FDH31	12.000	Regular	0.020407	Meat	99.9042	OUT035	2004	Small	
8514	FDA01	15.000	Regular	0.054489	Canned	57.5904	OUT045	2002	NaN	
8516	NCJ19	18.600	Low Fat	0.118661	Others	58.7588	OUT018	2009	Medium	
8520	NCJ29	10.600	Low Fat	0.035186	Health and Hygiene	85.1224	OUT035	2004	Small	
8522	DRG01	14.800	Low Fat	0.044878	Soft Drinks	75.4670	OUT046	1997	Small	

2439 rows × 12 columns

`Select the rows where Outlet_Location_Type is Tier 1 and Outlet_Establishment_Year is from the list [1987, 1988, 2009]`

In [5]:

```
# WRITE YOUR CODE HERE
filter1 = data.Outlet_Location_Type=='Tier 1'
filter2 = data.Outlet_Establishment_Year.isin([1987, 1988, 2009])
data[filter1 & filter2]
```

Out[5]:

Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Locat

Update the values of the column Outlet_Type using the mapping given below

In [6]:

```
mapping = {
    'Supermarket Type1' : 1,
    'Supermarket Type2' : 2,
    'Grocery Store'     : 3,
    'Supermarket Type3' : 4
}
```

In [7]:

```
# WRITE YOUR CODE HERE
data.Outlet_Type = data.Outlet_Type.map(mapping)
data.Outlet_Type.to_frame()
```

Out[7]:

	Outlet_Type
0	1
1	2
2	1
3	3
4	1
...	...
8518	1
8519	1
8520	1
8521	2
8522	1

8523 rows × 1 columns

Fill the missing values in the column 'Outlet_Size' by Medium using loc

In [8]:

```
data.isna().sum()
```

Out[8]:

```
Item_Identifier                 0
Item_Weight                  1463
Item_Fat_Content                0
Item_Visibility                 0
Item_Type                       0
Item_MRP                        0
Outlet_Identifier               0
Outlet_Establishment_Year       0
Outlet_Size                  2410
Outlet_Location_Type            0
Outlet_Type                     0
Item_Outlet_Sales               0
dtype: int64
```

In [9]:

```
# WRITE YOUR CODE HERE
fill_value = data.Outlet_Size.mode()[0]
data['Outlet_Size'].fillna(fill_value, axis=0, inplace=True)
data.Outlet_Size.unique()
```

Out[9]:

```
array(['Medium', 'High', 'Small'], dtype=object)
```